



Observation of Typhoon Trami (2024)-induced energy cascade from near-inertial waves to diurnal internal tides

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ABSTRACT

Energy transfers among internal waves in the northern South China Sea are not well characterized, particularly during typhoons, owing to the lack of in situ observations. Based on high-resolution mooring data collected during Typhoon Trami (2024), this study reveals the occurrence of robust vertical energy redistribution among diurnal internal tides (D1 ITs) and near-inertial waves (NIWs). Strikingly, the typhoon not only amplified the NIW energy but also triggered an unexpected surge in the D1 IT energy. The observed average net energy transfer rate of $1 \times 10^{-7} \text{ W kg}^{-1}$ from typhoon-forced NIWs to D1 ITs occurred at water depths of 120–170 m. Further bispectral analysis indicated that the energy transfer is driven by nonlinear wave–wave interaction. These results reveal the existence of a new energy transfer pathway—from atmospheric forcing to D1 ITs—and redefine the redistribution of the internal wave energy during extreme weather events.

摘要

台风期间内波间的能量传递特征及机制尚未得到清晰揭示。本研究利用 2024 年南海北部“潭美”台风期间获取的高分辨率潜标观测数据,发现全日内潮与近惯性内波之间存在显著的垂直能量串级现象。台风作用下,观测发现在近惯性波能量显著增强的同时,全日内潮能量亦出现激增;在 120–170 米深度范围的水层,观测到台风强迫下近惯性波向全日内潮的平均净能量传递率达 $1 \times 10^{-7} \text{ W kg}^{-1}$ 。双谱分析结果进一步证实,这一能量传递过程由非线性波-波相互作用主导驱动。上述结果揭示了“大气强迫至全日内潮”这一全新的内波能量传递路径,进而重新界定了极端天气事件作用下内波能量的再分配规律。

1. Introduction

Tropical cyclones (TCs) play a vital role in upper ocean dynamics by generating near-inertial waves (NIWs) through strong wind forcing. The South China Sea (SCS), as the largest marginal sea in the western North Pacific, experiences an average of approximately 10.2 typhoons annually (Wang et al., 2007). The NIWs induced by typhoons primarily dissipate through downward propagation as internal wave packets into the thermocline and deeper ocean layers (D'Asaro, 1989). Besides, the vertical shear associated with NIWs induces turbulent mixing through shear instability, resulting in the entrainment of colder thermocline water into the mixed layer and a reduction in sea surface temperatures by 1°C – 6°C along cyclone trajectories (D'Asaro et al., 2007). Furthermore, the downward propagation of near-inertial energy, influenced by background flows, enhances vertical mixing in deeper layers, facilitating energy cascades, turbulent dissipation, and the maintenance of large-scale ocean circulation (Chen et al., 2024). However, how NIWs affect the multiscale dynamics of oceanic internal waves requires further analysis.

The northern SCS is characterized by shallow continental shelves where powerful barotropic tides interact with uneven terrain, resulting in the generation of internal tides that exhibit varying resonance as they propagate (Xu et al., 2016). As these diurnal internal tides (D1 ITs) and semidiurnal internal tides (D2 ITs) propagate westward, they undergo nonlinear interactions with other internal wave components and other tidal constituents. These interactions can lead to the generation of higher-order harmonics and energy transfer between different frequency components (Xie et al., 2011). Besides, the energy dissipation usually results from the wave–wave interactions (WWI) and interactions with the bottom boundary, which subsequently influences the vertical stratification (Chen et al., 2024), modulates regional water mass exchange (Wang et al., 2017), and contributes to broader climate dynamics by redistributing heat and nutrients (Jayne, 2009). Numerous studies have indicated diverse energy transfer processes, yet energy cascades under typhoons remain uncertain.

Near-critical latitudes associated with the M2 and M1 tidal constituents have been shown to enhance the transfer of tidal energy

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into near-inertial motions, through resonant WWI, known as PSI (parametric subharmonic instability) (Hu et al., 2023; Yang et al., 2018, 2020). Typhoons significantly amplify energy transfer rates to high-frequency internal waves (HFIWs), with storm-induced inertial oscillations altering the kinetic to baroclinic energy transfer (Sun and Pinkel, 2012; Yang et al., 2022, 2024). An observational study conducted by Guan et al. (2014) in the northern SCS revealed the occurrence of increased energy transfer rates between NIWs and D1 ITs following typhoon occurrences. This research emphasized that wave-wave non-linear interactions and the burst of surface diurnal internal tides were attributed to stratification changes and wind-generated currents. However, due to the resolution limitations of observations, many studies have been unable to detect changes in deep-layer internal tides, have overlooked the energy transfer from NIWs to D1 ITs, and have lacked further quantification of the transfer efficiency during typhoons.

Understanding the energy cascade during typhoon events is pivotal for directing internal tide energy towards small-scale turbulent dissipation, while accurately estimating these transfer rates is critical for understanding regional energy budgets (Sun and Pinkel, 2013), both of which would aid in refining turbulent mixing parameterization for intricate internal tide scenarios in the SCS. Hence, there is an urgent need to clarify the characteristics and mechanisms of this energy cascade during typhoons, particularly regarding how TCs influence energy transfer between NIWs and D1 ITs. Our study elucidates NIW-D1 IT energy transfer during extreme events like typhoons, quantifies the magnitude of energy transfer, and further discusses the mechanisms. We utilize distinctive high-resolution mooring data collected during Typhoon Trami (2024) in the northern SCS and demonstrate a significant relationship between the NIWs generated by the typhoon and the surrounding internal tide field. Furthermore, analysis of the energy transfer dynamics between near-inertial motions and internal tides is conducted. The results of this study elucidate the energy transfer between NIWs and D1 ITs during extreme weather events such as typhoons, advancing our understanding of cross-frequency energy cascade dynamics in oceanic systems. This understanding will contribute to the refinement of turbulent mixing parameterization in intricate internal tide situations within the SCS.

2. Data and methods

2.1. Data

A high temporal resolution mooring was deployed in the northern SCS at (17.45°N, 112.21°E) (red pentagram in Fig. 1(a)), with a bottom depth of 1365 m. It operated continuously from 20 October 2024 to 15 December 2024. It was equipped with an upward-looking 75 kHz acoustic Doppler current profiler (ADCP) (located 500 m beneath the sea surface) to measure current velocities and a thermistor chain consisting of 21 conductivity–temperature–depth (CTD) or temperature–depth (TD) sensors at varying depths. For further details, refer to Table S1 and Fig. S1.

The TC best-track data from the China Meteorological Administration (CMA) provided six-hourly updates on the typhoon center's location and intensity. The 10-m wind speed and direction time series were extracted from ERA5 (fifth generation ECMWF atmospheric reanalysis) data, which contain hourly gridded meteorological parameters (Fig. S2). Barotropic tidal current data at the mooring location were predicted from the TPX09 dataset. Calculations of sea surface vorticity were conducted using sea level anomaly and geostrophic currents obtained from the Copernicus Marine Data Service.

2.2. Methodology

To characterize the current variations at different frequencies, we utilized a third-order Butterworth filter across the entire time domain for the observations. After subtracting the barotropic current from

the TPX09 model, the remaining current variations were divided into $[(0.80, 1.20) f]$, $1/(26\text{--}22 \text{ h})$, and $1/(14\text{--}10 \text{ h})$, respectively, where f denotes the local inertial frequency. The horizontal kinetic energy (HKE) was calculated using the formula $\text{HKE} = \frac{1}{2}\rho_0(u'^2 + v'^2)$ (u' and v' represent residual currents), with a constant seawater density of $\rho_0 = 1024 \text{ kg m}^{-3}$. The HKEs in the near-inertial and diurnal frequency bands were computed.

3. Results

3.1. Vertical-temporal variability

An 11-day depth–time analysis (22 October to 2 November) was conducted, incorporating three-day high-pass and six-day low-pass filtered meridional velocities, as well as vertical shear of raw velocity (computed as $S^2 = (\partial u/\partial z)^2 + (\partial v/\partial z)^2$) and temperature variations (Fig. 1(b–e)). Key changes in sub-three-day internal wave dynamics were observed following the typhoon: after 27 October, velocity amplitudes increased, with surface-layer peaks reaching approximately 0.4 m s^{-1} . This contrasted with pre-27 October conditions, characterized by lower velocities ($\sim 0.2 \text{ m s}^{-1}$) and a primarily diurnal period—consistent with previously reported features of internal tides in the northern SCS (Chen et al., 2024). Additionally, two positive velocity peaks on 28 and 30 October exhibited distinct periods: $\sim 40 \text{ h}$ in waters shallower than 140 m and $\sim 30 \text{ h}$ in deeper waters ($>140 \text{ m}$), suggesting interactions between internal waves across different frequency bands (Fig. 1(b)).

Low-frequency waves (with frequency below 6 cycles per day (cpd)) also showed typhoon-induced changes, with velocity amplitudes rising from $<0.05 \text{ m s}^{-1}$ before 26 October to around 0.1 m s^{-1} by 28 October. Additionally, the result also illustrates that the mooring location was not affected by subsurface eddies prior to the typhoon, and the strengthening of large-scale background currents after 26 October was attributed to the typhoon (Fig. 1(c)). Concurrently, vertical shear in the upper 100 m of the water column increased significantly following the typhoon (Fig. 1(d)). This increase indicates changes in shear instability driven by intense winds, which likely enhanced mixing in the upper ocean layers.

In addition, the temperature variability mirrors the high-frequency current behavior: before the typhoon, the temperatures were primarily influenced by D1 ITs, while the near-inertial oscillations caused by the typhoon produced temperature changes that exhibited the characteristics of inertial periods (Fig. 1(e)). In addition, the mixing and entrainment, along with the upwelling of cold water from the thermocline caused by the typhoon, resulted in a noticeable elevation of the isotherms in the upper layers (Price, 1981). Thus, following the passage of the typhoon, the seawater temperature at 50 m decreased by approximately 4°C .

3.2. Impact of the typhoon on the internal wave energy

Under the influence of the typhoon, the local HKE of the NIWs at the mooring location increased rapidly and propagated to deeper layers over time. Two peak depths of the near-inertial energy were concentrated at water depths of $<100 \text{ m}$ (values $>10 \text{ J m}^{-3}$) and at a depth of 250 m (6.5 J m^{-3}). The deeper high-value center and the shallower high-value center emerged almost simultaneously due to the impact of the typhoon (Fig. 2(a)). In addition, the time of the diurnal tidal peak energy coincided with the peak time of the near-surface near-inertial energy, with a central peak, which also exceeded 10 J m^{-3} , occurring at depths of 140–180 m, which was almost consistent with the depth where the near-inertial energy exhibited weak values (Fig. 2(a, b)). Based on the timing of the diurnal barotropic tides (K1+O1) obtained from TPX09, the peaks of the diurnal tidal energy on 22 October and 4 November were due to the barotropic spring tides, as the peaks in the D1 ITs occur from 27 October to 30 October (Fig. 2(b, c)). In addition, the coherent ITs propagating from remote areas have little impact on

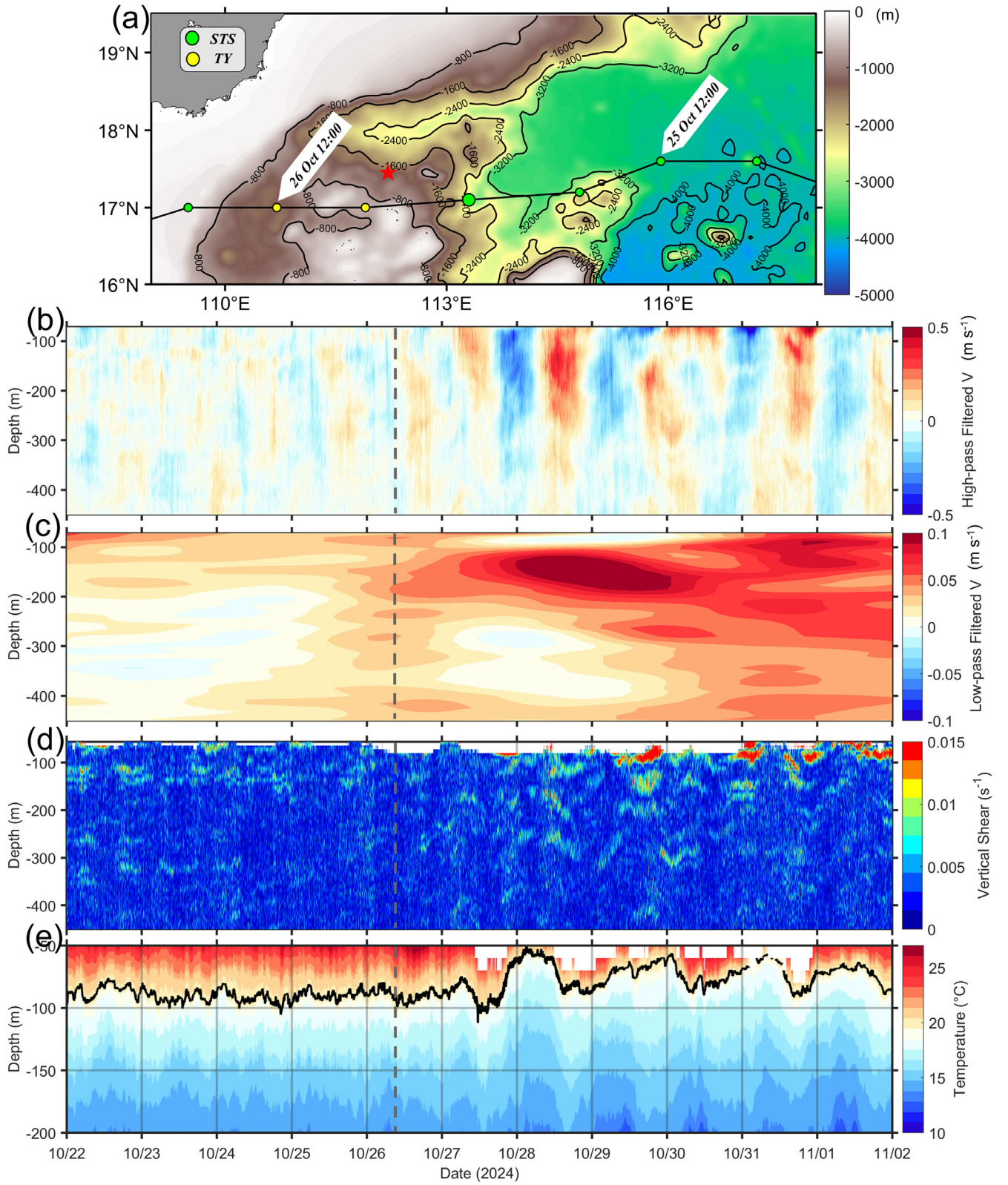


Fig. 1. (a) The mooring position in the southern SCS (red pentagram). The background color and contour lines represent the water depth. Eleven-day time–depth maps of (b) high-pass filtered meridional velocity below a three-day period, (c) low-pass filtered meridional velocity over a six-day period, (d) vertical shear of the raw meridional velocity, and (e) temperature. The solid black line in (e) denotes the 20 $^{\circ}\text{C}$ isotherm. The vertical gray dashed line denotes the time when the typhoon was closest to the mooring station.

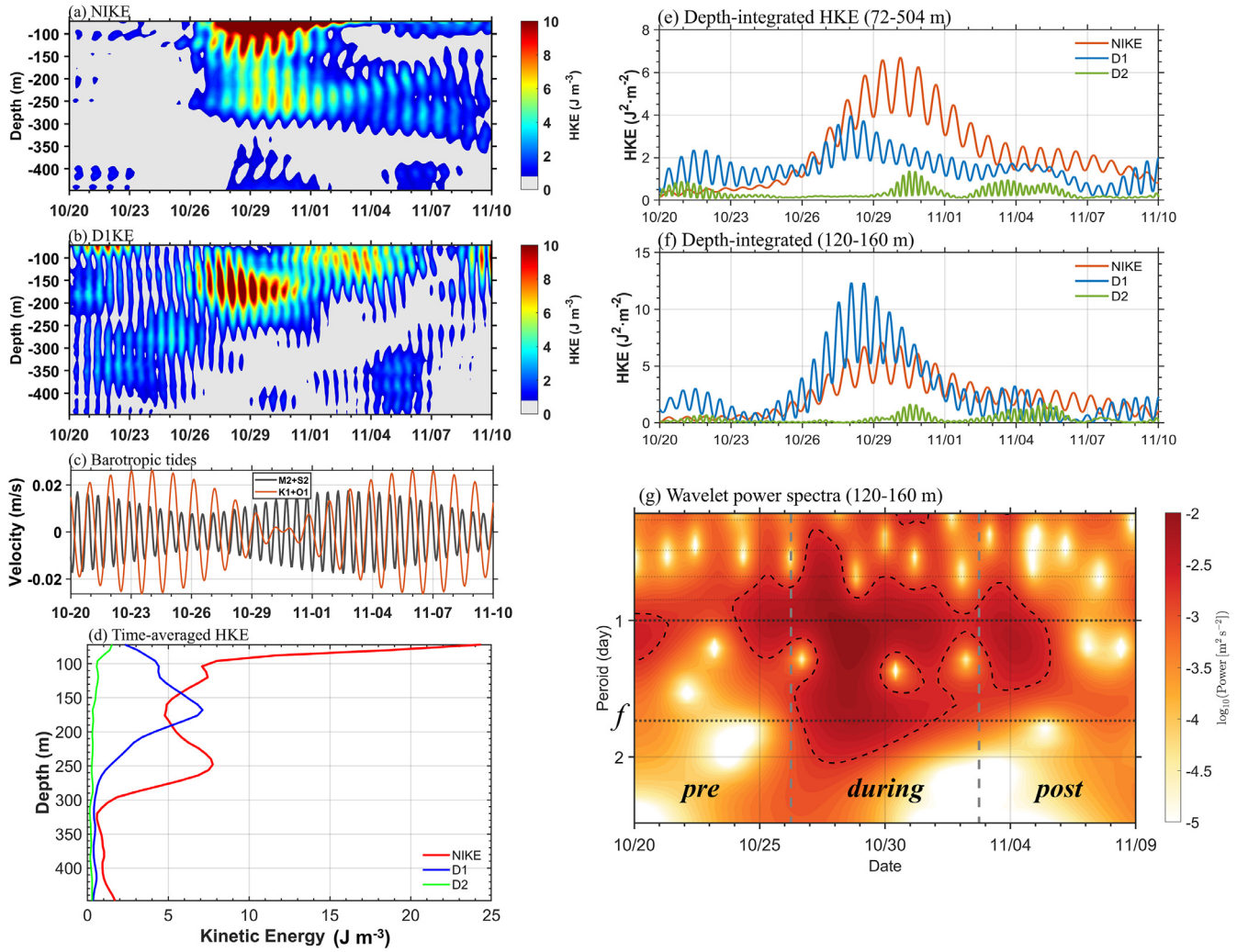


Fig. 2. The vertical HKE distributions of the (a) near-inertial motions and (b) D1 ITs. (c) The amplitudes of the semidiurnal (M2+S2) and diurnal (K1+O1) barotropic tidal currents at the mooring station as predicted from TPXO9. (d) Depth profiles of the time-averaged HKE during the typhoon. Time series of the vertically depth-integrated HKE (e) from 72 to 504 m and (f) from 120 to 160 m. (g) Wavelet power spectra of the meridional velocity at depth of 120–160 m. The gray dashed line indicates the typhoon arrival time, and the black dashed line represents the 95 % significance threshold.

the local D1 ITs (Fig. S3). This further demonstrates that the typhoon significantly enhanced the diurnal internal wave energy at the mooring location.

Analysis of the wavelet spectrum at middle depths (120–160 m) revealed that following the passage of the typhoon on 26 October, there was a significant concentration of the peak power spectral density within the near-inertial and diurnal tidal frequency bands, while there was some enhancement in the other frequency bands. The enhanced near-inertial HKE peak was sustained for approximately seven days, whereas the diurnal tidal peak lasted longer (Fig. 2(g)). This finding suggests that the near-inertial HKE serves as the predominant dynamic process in the upper ocean in the aftermath of a typhoon, corroborating the results of previous research (Zhang et al., 2022).

Analysis of depth-integrated HKE across different depth ranges elucidated the distribution of energy across frequencies and depths (Fig. 2(e, f)). Within the entire 72–504 m layer, the near-inertial HKE was the most significant during the typhoon period. Conversely, in the 120–160 m layer, the HKE of the D1 ITs was the most prominent. As shown in Fig. 2(e), the near-inertial energy peaks at 6.5 J m^{-3} , whereas Fig. 2(f) shows a diurnal tidal peak energy of 7 J m^{-3} . In addition, the time-average near-inertial HKE decreased sharply above a depth of 160 m. By contrast, the HKE of the D1 ITs remained relatively stable

at lower values (averaging around 3 J m^{-3}) in the shallow layer above 120 m, increased slightly from 120 to 170 m, and peaked at 8 J m^{-3} at 170 m, with these energy distribution characteristics indicating an inverse relationship between the two energy components in the 120–170 m depth range (Fig. 2(d)).

3.3. Energy transfer from NIWs to D1 ITs

The energy transfer process was calculated using Eq. (S5), with the detailed derivation of the equation and additional analysis provided in Text S1–S2 and Fig. S4. During the typhoon event, a notable alteration of the computed value was observed, indicating substantial energy transfer between the two components of the internal waves (Fig. 3). A notable positive peak of around $2 \times 10^{-7} \text{ W kg}^{-1}$ was observed within the temporal range of the typhoon's influence, specifically spanning depths of 120–170 m, from 27 October to 31 October. This period aligns closely with the depth and timing of the diurnal tidal energy enhancement phenomenon evident in the energy distribution (Fig. 3(a)). Concurrently, depth-averaged integration of the energy transfer rate within the 120–170 m depth was conducted. It was found that an average energy transfer rate of around $1 \times 10^{-7} \text{ W kg}^{-1}$ occurred between 1200 UTC 27 October and 1200 UTC 31 October (Fig. 3(b)).

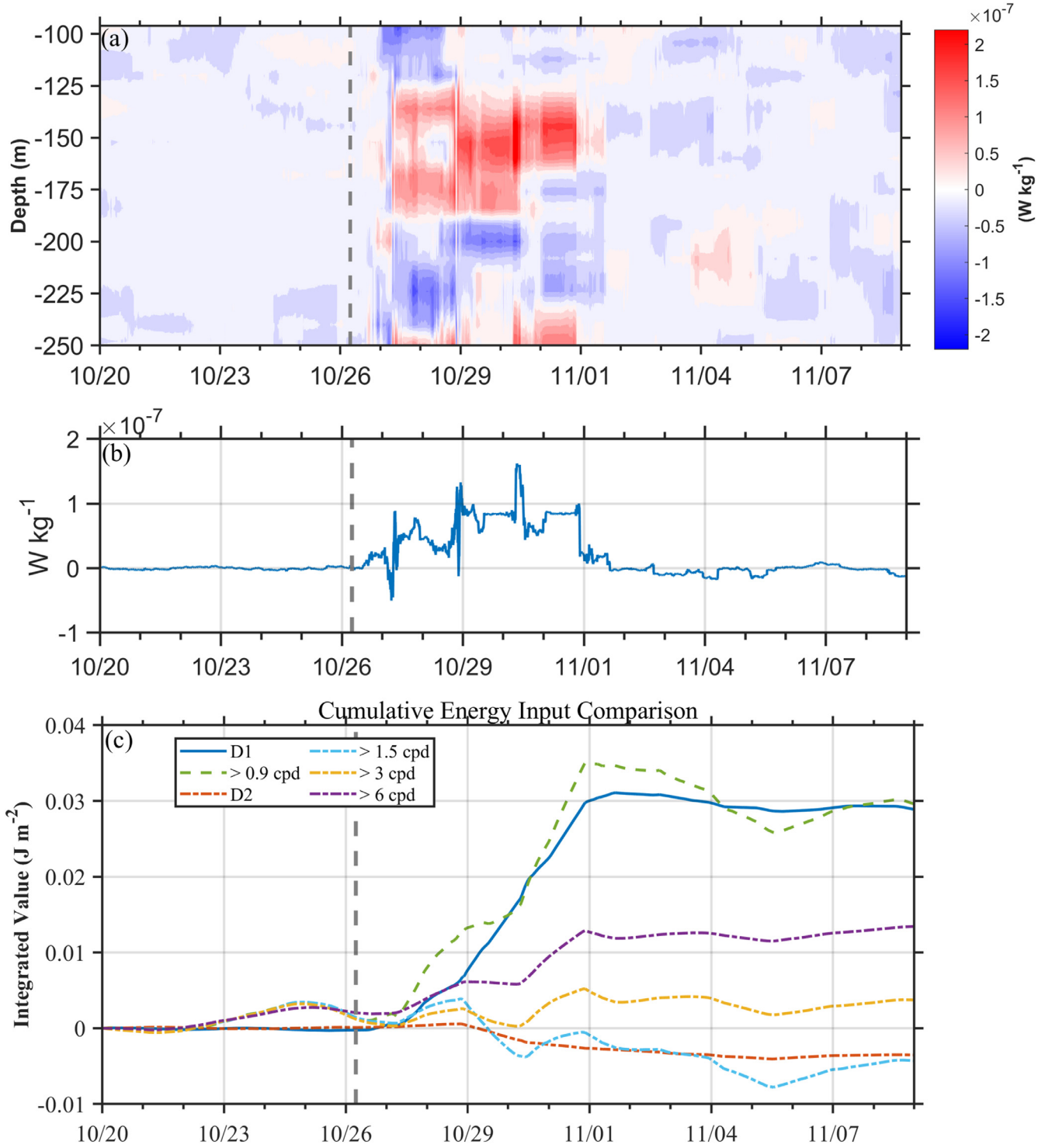


Fig. 3. (a) Energy transfer rate (units: W kg^{-1}) obtained using Eq. (S5), and (b) corresponding depth-averaged values from 120 to 170 m (units: W kg^{-1}). (c) The energy transfer integral input derived by calculating the HFIW in different frequency ranges. D1 and D2 ITs were obtained via third-order Butterworth band-pass filtering with bandwidths of $[1/26, 1/22]$ h, and $[1/14, 1/10]$ h, respectively. The vertical gray dashed line denotes the time when the typhoon was closest to the mooring station.

Furthermore, the cumulative energy input (EI; J m^{-2}) from the near-inertial motions to the tidal forces over a specified time interval ($t_1 - t_2$) throughout the water column (0–H) was calculated using Eq. (S6). The findings indicate that, induced by the typhoon, the maximum energy accumulation resulting from the passage of the typhoon transferred $3.2 \times 10^{-2} \text{ J m}^{-2}$, followed by a slight decrease. It ultimately stabilized

at $3.0 \times 10^{-2} \text{ J m}^{-2}$ (solid blue line in Fig. 3(c)). Moreover, based on an examination of the different pre-filtering frequency ranges, it was found that there was a moderate transfer of near-inertial energy to other high-frequency bands. Notably, the amplification of the diurnal tidal energy was especially significant (Fig. 3(c)). The results demonstrate the energy cascade of the near-inertial energy to the diurnal tidal energy, empha-

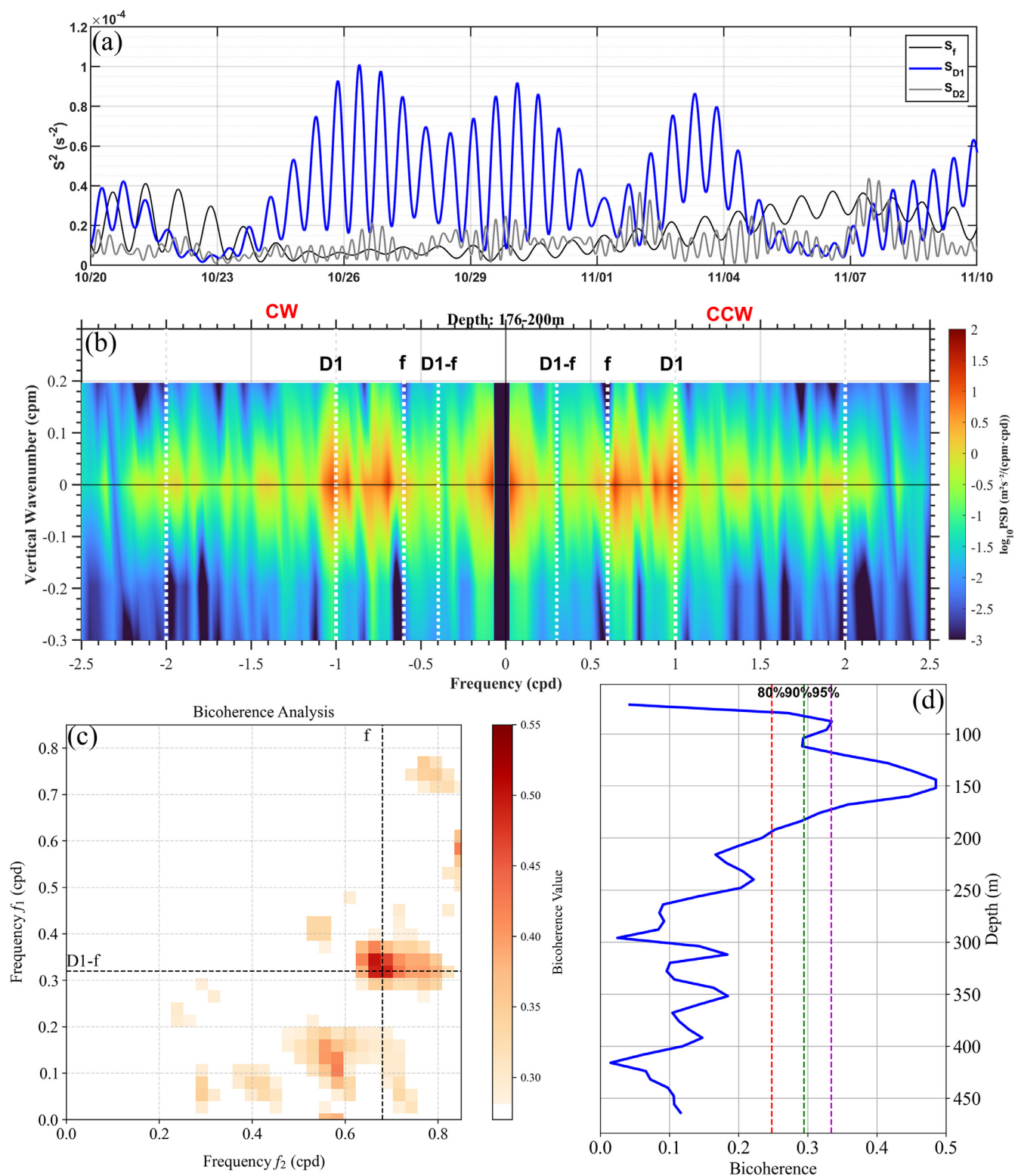


Fig. 4. (a) Time series of vertical integration of f , D1, and D2 shears between 176 and 200 m. (b) Wavenumber–frequency spectra of 176–200 m. (c) Mean bicoherence values of u and v velocity component series at 152 m depth during 27 October to 6 November. (d) Depth variations of bicoherence values for meridional velocity during the typhoon period (26 October to 6 November) at [0.68, 0.32] cpd. Vertical dashed lines from left to right represent significance level thresholds at 0.2, 0.1, and 0.05, respectively.

sizing that the typhoon had an important impact on the redistribution of internal wave energy in the interior of the ocean.

3.4. Typhoon-triggered enhanced nonlinear WWI

Nonlinear WWI, is a resonant WWI process that plays a crucial role in the transfer of energy from larger parent waves, such as D1 and D2 ITs, to smaller subharmonic daughter waves (Hazewinkel and Winters, 2011; Müller et al., 1986). Typically, nonlinear WWI involves the transfer of energy from waves with lower vertical modes to those with higher vertical modes, necessitating a low-wave-number wave to provide energy and a high-wave-number wave to absorb it. In general, higher wave numbers are linked to greater velocity shear (Richet et al., 2017). The shear and wavenumber-frequency spectra were analyzed to identify potential wave-wave energy transfer processes. Fig. 4(a) shows that, during the typhoon's passage (26 October to 4 November), D1 ITs exhibited the strongest shears within the 176–200 m depth range, with a depth-integrated average of approximately $1 \times 10^{-4} \text{ s}^{-2}$. In contrast, NIWs and D2 ITs had weaker shears, with depth-integrated magnitudes of $\sim 2 \times 10^{-5} \text{ s}^{-2}$ (Fig. 4(a)). Furthermore, a comparison of the wavenumber–frequency spectra for the two polarization directions revealed that D1 ITs exhibited the highest wavenumbers, followed by NIWs, while D2 ITs displayed the lowest wavenumbers (Fig. 4(b)).

Bicoherence values were further calculated to reveal nonlinear WWI between D1 ITs and NIWs during the occurrence of a typhoon. A prominent bicoherence peak was identified at the frequency coordinate [0.68, 0.32] cpd at a depth of 152 m (Fig. 4(c)). Concurrently, vorticity calculations at the mooring location revealed that the 0.68 cpd frequency corresponds precisely to the local effective inertial frequency at 152 m (Fig. S5 and S6). Therefore, nonlinear WWI occurred within a triad system comprising f , D1- f , and D1 during typhoon activity. Such a coupling relationship, alongside significant nonlinear WWI among D1- f waves, NIWs, and D1 waves, has also been documented in the northern SCS, the East China Sea, and the North Pacific (Carter and Gregg, 2006; Xie et al., 2016; Xu et al., 2022; Yang et al., 2020). Moreover, the depth profiles of bicoherence values revealed notable peaks within the depth range of 150 to 170 m, while neither the shallow layer (<150 m) nor the deep layer (greater than 200 m) exceeded the significance threshold (Fig. 4(d)). This specific depth range aligns with the depth of vertical peak diurnal tidal energy and weak peak NIWs (Fig. 2 (a, b)). In summary, the evidence of nonlinear WWI and the energy transfer rate collectively confirm the direction of energy transfer from NIWs to D1 ITs, which occurs at depths of 150 and 200 m.

4. Discussion and conclusions

Previous research has highlighted key aspects of typhoon-related internal wave energy transfer. Guan et al. (2014) and Yang et al. (2024) both observed the transfer of near-inertial energy to high-frequency energy during typhoons, with the latter quantifying a transfer rate of $2 \times 10^{-6} \text{ W kg}^{-1}$. Gong et al. (2024) used modeling to link typhoon-induced NIW-to-internal tide (IT) energy transfer to the energy pipeline effect. Additionally, Zhang et al. (2025) showed that resonant wave interactions facilitate KE exchange between NIWs and ITs. Collectively, these studies indicate that internal wave energy cascades during typhoons are active yet complex, with many underlying mechanisms still requiring further investigation.

This study analyzed mooring observations to characterize post-typhoon variations in internal tide energy within the northern SCS, shedding light on the energy transfer mechanisms triggered by storm events. Results show that typhoon-energized NIWs serve as a critical energy source, driving energy transfer to D1 ITs at specific depths. Key findings reveal depth-dependent shifts in dominant internal wave energy: NIW energy dominates the upper ocean but decreases abruptly below the surface layer, while D1 IT energy intensifies progressively with depth. A distinct seesaw relationship between these energy components emerged

between 120 and 170 m, and the diurnal tide energy peaks occurred at 170 m. This diurnal tide energy peak is caused neither by barotropic tide variations nor by changes in stratification (see Fig. 2, Text S2, and Fig. S4). Notably, during the typhoon, the average energy transfer rate at 120–170 m reached approximately $1 \times 10^{-7} \text{ W kg}^{-1}$, playing a pivotal role in modulating this energy cascade and facilitating energy redistribution across the water column. Further bispectral analysis indicates that the diurnal tide energy peak primarily arises from near-inertial energy transfer via the nonlinear WWI. Typhoons significantly modulate internal wave energy transfer by enhancing nonlinear WWI between tidal and inertial frequency bands, thereby facilitating oceanic energy redistribution.

These findings challenge the conventional framework of internal wave energy dynamics by underscoring the pivotal role of typhoon forcing in reshaping spectral energy distributions. Our analysis reveals that pronounced energy enhancement in the diurnal tidal band occurs concurrently with the mid-depth attenuation of NIWs, suggesting a direct energy cascade from NIWs to tidal frequencies. Consequently, improved representation of typhoon-driven energy cascade processes in ocean models is critical for improving the accuracy of mixing parameterizations, particularly under extreme weather scenarios. In addition, WWIs and energy transfer among internal waves are modulated by background oceanic flow, such as mesoscale eddies (Xu et al., 2022). Another notable observation is that typhoon-induced negative energy transfer occurs from NIWs to the D2 frequency band, which in turn results in negative energy transfer to waves with frequencies >1.5 cpd (see red and blue dashed lines in Fig. 3(c)). This may also stem from three-wave nonlinear interactions, though the mechanism requires further study. The finding confirms that intense typhoon-driven NIW injection into the northern SCS induces strong nonlinear interactions with pre-existing internal tides, significantly impacting oceanic energy cascades.

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Data availability statement

The daily geostrophic currents are accessible at <http://marine.copernicus.eu>. The barotropic tidal currents were obtained from TPXO9 (<https://www.tpxo.net/tpxo-products-and-registration>). The ERA5 wind data were obtained from <https://cds.climate.copernicus.eu/datasets>, and the GEBCO bathymetry data were obtained from <https://www.gebco.net/data-products/gridded-bathymetry-data>. The CMA TC best-track data were accessed at <https://tcdata.typhoon.org.cn/en/>. The CTD and ADCP mooring data analyzed can be downloaded from <https://doi.org/10.6084/m9.figshare.29276060>.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.aosl.2025.100712](https://doi.org/10.1016/j.aosl.2025.100712).

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